HRT Challenge - How to solve a problem in 2026

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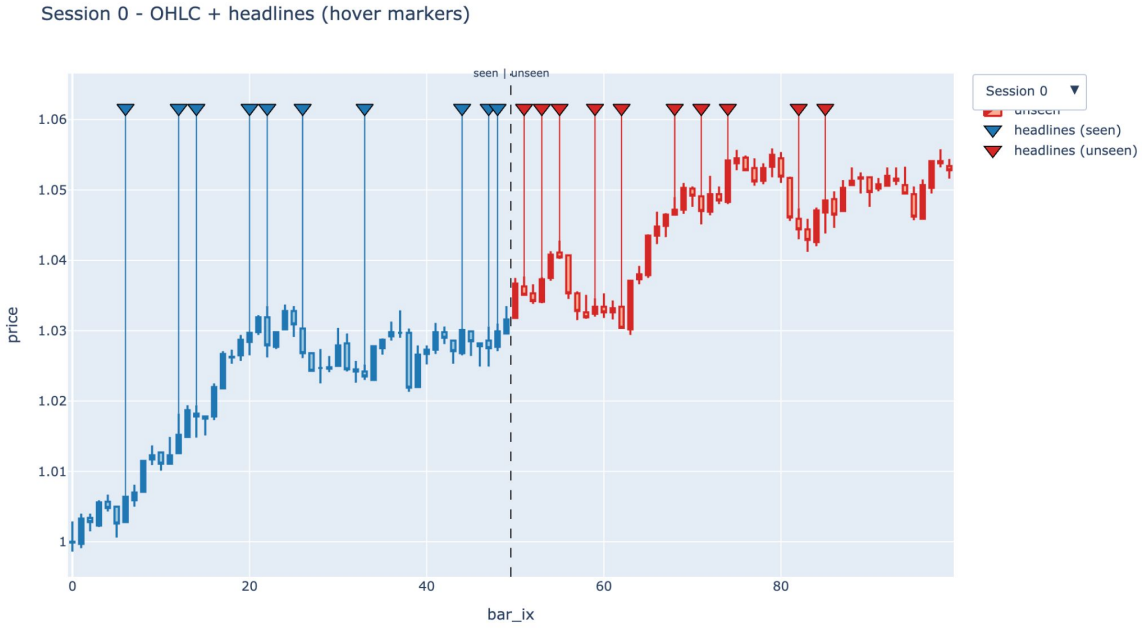
Problem Statement



Fixed Templates as Newsheadlines

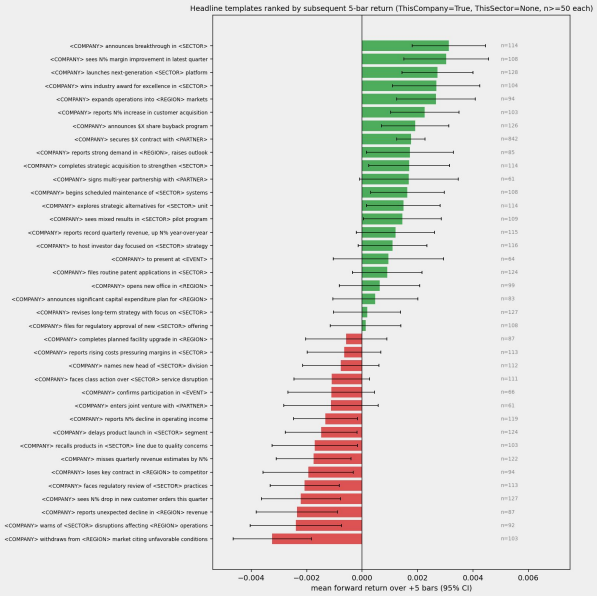
- Headlines limited to a handful of Templates
- Effect of templates heavily correlated within groups

→ We build a Parser that parses headlines into one of 49 templates

```
<COMPANY> <ROLE> steps down unexpectedly citing <REASON>
```

```
<COMPANY> faces regulatory review of <SECTOR> practices
```

- No real success with specific companies/sectors
- No real success with checking difference in amplitude of individual headlines

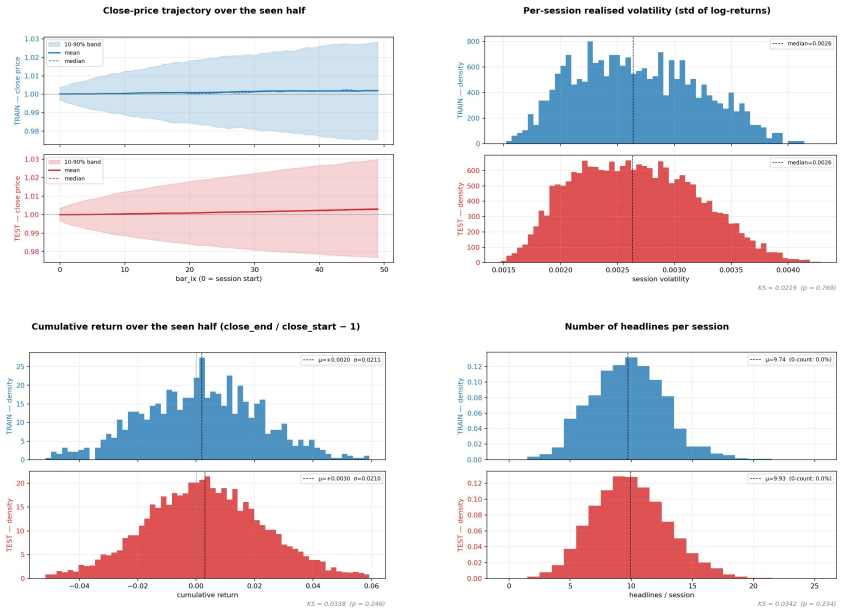


No real difference/drift between train and test

- We found no difference between train and test data
- train data size 1000, test data size 20 000

→ we thought of and tried utilizing test data to help train our model, w.o. a lot of success

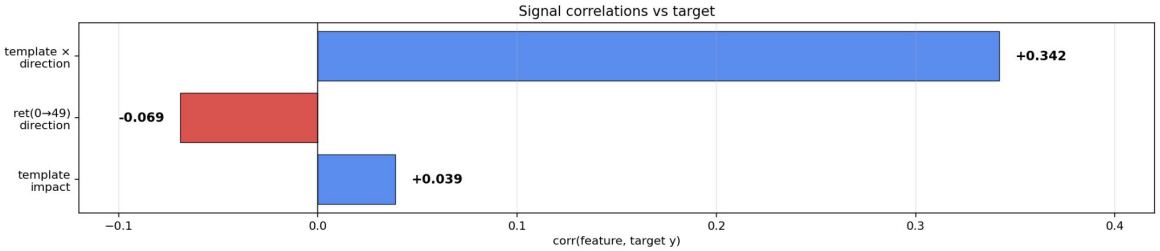
- Different approaches, GRN to pretrain bar structure on 20 000
- CatBoost trained on predicting less than 50 into the future on test then generalizing



Reality Check

- After 8 hours of optimizing our models we hit a plateau
- At this point we had pre-selection feature set of around 200, yet most our models where basically “just going long”
- Hypothesis ⟶ There just is not that much signal in the data

⟶ We tested with a single regression that includes only mean return per headline template, volatility and mean cum_return after 50 candles, This blew all our other more sophisticated models out of the water

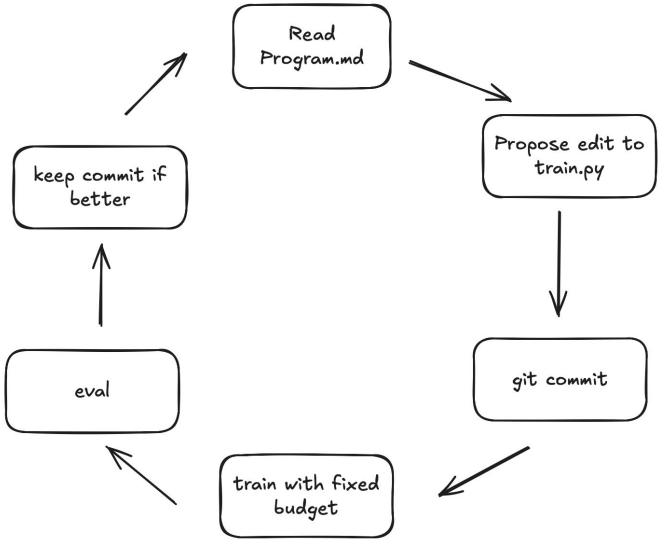


Auto research Framework

- published by Andrew Karpathy recently
- base idea recurrent research loop with llm
- start at strong baseline then iteratively change small elements and check back

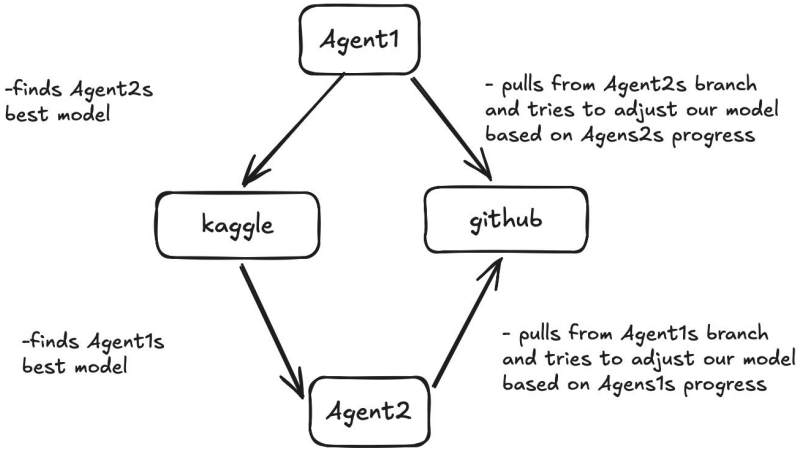
→ Ideal setup for us, start with the simple linear regression and iteratively build up

One training loop every ~ 8 min



Our agent implementation

- We had two agents run this loop concurrently
- shared github, exchanged best solutions after each training loop
- wired each agent to own Kaggle account to make commits and receive public test score



Final Implementation

Signal construction (core model): Blend a strong base model (“champ”) with a per-headline ridge ensemble (multi-α, interaction-rich, short horizon K=3), plus a low-correlation session-level stack (`stk`) to inject orthogonal signal.

Calibration & combination: Apply a small negative shift to the main signal (reduces variance), then combine weighted components into a single prediction:
main blend ≈ `0.91*(headline signal) + 0.12*(stack)`.

Risk scaling (key edge): Normalize predictions by session volatility using **Parkinson vol** with exponent ~1.2 — this downweights noisy sessions and delivers the biggest performance gain.